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Project Report

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The Internet of Things (IoT) is a rapidly growing area in research, with a wide range of applications in various fields. As a result, so many challenges arise in both device management and security of these devices. So, it is necessary to have a high-quality IoT profiling dataset. Which is done by the authors of “Towards the Development of a Realistic Multidimensional IoT Profiling Dataset”. In this paper, the behavior of 60 IoT devices was analyzed while experiments were conducted at Canadian Institute for Cybersecurity. The devices are WiFi, ZigBee, and Z-Wave devices. The data was collected from each device in four stages. Such as power on, power off, active, and interactions. The dataset is designed to give a realistic idea of IoT devices and their behavior.

After investigating the datasets, we came to know that, all the information was captured using Wireshark. Which is an open-source network traffic analyzer, used by millions of networking professionals. Another tool that was also used to capture data is “Dumpcap”. Which is used for semi-automated tests. The data was collected for all connected devices. One is when the device is on another is when the device is off or idle. There are a total of three types of devices. Audio, Camera, and Home Automation. Among these, the majority of devices are cameras which are about 84% of the data, and the proportion for Audio and Home automation devices was about 8% each. A total of 60 IoT devices were used to collect the data for the Dataset. Such as Cameras, Smart lamps, Speakers, Smart hubs, Multisensor, actuators, Motion sensors, etc. Different devices from different brands were used to capture the data. By conducting experiments in different phases, this dataset is prepared with 48 extracted features.

In our Mini Project, we had to perform some preprocessing steps on this dataset. In the provided dataset directory, we got 3 additional directories named, Audio (Including 126 CSV files), Camera (Including 162 CSV files), and Home Automation (Including 363 CSV files). After that, we merged each CSV file from each directory and got three CSV files named 'merged_audio', 'merged_camera', and 'merged_homeAuto'. After that, we performed some feature engineering. For Audio, Camera, and Home Automation, in each CSV file, we have added two extra features named 'Device_Name' and 'Device_Number' (Since we need numeric values to train and test our model). After that, by merging these three files, we produced our final dataset named 'merged_df' consisting of 229757 rows and 51 features (With 2 additional features) and continued with our further analysis.

Here are the codes and outputs of our analysis,

```python
import os
folder_path = '/content/drive/MyDrive/Colab Notebooks/Audio'

df_list = []

for file_name in os.listdir(folder_path):
    if file_name.endswith('.csv'):
        df = pd.read_csv(os.path.join(folder_path, file_name))
        df_list.append(df)

merged_audio = pd.concat(df_list, ignore_index=True)
merged_audio['Device_Type'] = 'Audio'
merged_audio['Device_number'] = 1
merged_audio.to_csv('merged_audio.csv', index=False)

merged_audio
```

<u>Figure 01:</u> Producing 'merged_audio' dataset

	Unnamed: 0	L4_tcp	L4_udp	L7_http	L7_https	port_class_src	port_class_dst	pck_size	ip_dst_new	ethernet_frame_size	...
0	0	0	1	0	0	2	2	44	4026531835	78	...
1	1	0	1	0	0	2	2	44	3758096634	78	...
2	2	0	0	0	0	0	0	64	3232270593	98	...
3	3	0	1	0	0	2	1	40	134744072	74	...
4	4	0	1	0	0	1	2	56	3232270754	90	...
...	...	...	...	...	...	...	...	...	...	...	...
18769	42	1	0	0	1	1	2	66	3232270803	100	...
18770	43	1	0	0	1	2	1	20	2954913076	54	...
18771	44	1	0	0	1	2	1	66	2954913076	100	...
18772	45	1	0	0	1	1	2	20	3232270803	54	...
18773	46	1	0	0	1	1	3	20	3232270803	54	...

18774 rows × 51 columns

<u>Figure 02:</u> 'merged_audio' dataset (I)

...	kurt_e	var_e	q3_e	q1_e	iqr_e	epoch_timestamp	inter_arrival_time	time_since_previously_displayed_frame	Device_Type	Device_number
...	0.074898	1594.240000	44.0	32.0	12.0	1.634054e+09	0.000000	0.000000	Audio	1
...	0.858579	1336.555556	44.0	32.0	12.0	1.634054e+09	0.000000	0.000000	Audio	1
...	0.968490	1168.489796	54.0	32.0	22.0	1.634054e+09	0.025100	0.025100	Audio	1
...	1.705787	1038.937500	49.0	32.0	17.0	1.634054e+09	0.740150	0.765250	Audio	1
...	2.127407	926.222222	56.0	32.0	24.0	1.634054e+09	0.025906	0.791156	Audio	1
...	...	...	...	...	...	...	...	...	...	...
...	2.701632	1609.982222	58.0	20.0	38.0	1.633614e+09	3.378840	4.228406	Audio	1
...	3.127789	1540.714844	58.0	20.0	38.0	1.633614e+09	0.075152	4.303558	Audio	1
...	2.948828	1482.809689	58.0	20.0	38.0	1.633614e+09	0.000000	4.303558	Audio	1
...	3.321677	1428.472222	58.0	20.0	38.0	1.633614e+09	0.029215	4.332773	Audio	1
...	3.698726	1377.058172	58.0	20.0	38.0	1.633614e+09	8.409050	12.741823	Audio	1

<u>Figure 03:</u> ‘merged_audio’ dataset (II)

```
folder_path1 = '/content/drive/MyDrive/Colab Notebooks/Camera'

df_list1 = []

for file_name in os.listdir(folder_path1):
    if file_name.endswith('.csv'):
        df = pd.read_csv(os.path.join(folder_path1, file_name))
        df_list1.append(df)

merged_camera = pd.concat(df_list1, ignore_index=True)
merged_camera['Device_Type'] = 'Camera'
merged_camera['Device_number'] = 2
merged_camera.to_csv('merged_camera.csv', index=False)

merged_camera
```

<u>Figure 04:</u> Producing ‘merged_camra’ dataset

	Unnamed: 0	L4_tcp	L4_udp	L7_http	L7_https	port_class_src	port_class_dst	pck_size	ip_dst_new	ethernet_frame_size	...	kurt_e
0	0	0	1	0	0	2	2	72	3232270647	106	...	-1.411517
1	1	0	1	0	0	2	2	72	794308863	106	...	-1.290329
2	2	0	1	0	0	2	2	72	794308863	106	...	-1.106339
3	3	0	1	0	0	2	3	72	2859101655	106	...	-0.885321
4	4	0	1	0	0	3	2	144	3232270647	178	...	0.729717
...	...	...	...	...	...	...	...	...	...	...	...	...
191170	37	0	1	0	0	2	2	12	313968685	46	...	-1.721847
191171	38	0	1	0	0	2	2	12	64861754	46	...	-1.733241
191172	39	0	1	0	0	2	2	28	3232270632	62	...	-1.634305
191173	40	0	1	0	0	2	2	28	3232270632	62	...	-1.528450
191174	41	0	1	0	0	2	2	28	3232270632	62	...	-1.417238

191175 rows × 51 columns

<u>Figure 05:</u> 'merged_camera' dataset (I)

kurt_e	var_e	q3_e	q1_e	iqr_e	epoch_timestamp	inter_arrival_time	time_since_previously_displayed_frame	Device_Type	Device_number
-1.411517	596.244898	72.0	32.0	40.0	1.633541e+09	0.000000	0.000000	Camera	2
-1.290329	551.750000	72.0	36.0	36.0	1.633541e+09	0.003641	0.003641	Camera	2
-1.106339	511.209877	72.0	40.0	32.0	1.633541e+09	0.001242	0.004883	Camera	2
-0.885321	475.040000	72.0	46.0	26.0	1.633541e+09	0.035542	0.040425	Camera	2
0.729717	1009.454545	72.0	52.0	20.0	1.633541e+09	0.088222	0.128647	Camera	2
...	...	...	...	...	...	...	...	...	...
-1.721847	357.702479	56.0	20.0	36.0	1.633537e+09	0.000000	23.919327	Camera	2
-1.733241	373.916667	56.0	18.0	38.0	1.633537e+09	0.000000	23.919327	Camera	2
-1.634305	348.153846	56.0	20.0	36.0	1.633537e+09	0.029457	23.948784	Camera	2
-1.528450	325.673469	53.0	20.0	33.0	1.633537e+09	0.220400	24.169184	Camera	2
-1.417238	305.893333	50.0	20.0	30.0	1.633537e+09	0.037237	24.206421	Camera	2

<u>Figure 06:</u> 'merged_camera' dataset (II)

```python
folder_path2 = '/content/drive/MyDrive/Colab Notebooks/Home Automation'

df_list2 = []

for file_name in os.listdir(folder_path2):
    if file_name.endswith('.csv'):
        df = pd.read_csv(os.path.join(folder_path2, file_name))
        df_list2.append(df)

merged_homeAuto = pd.concat(df_list2, ignore_index=True)
merged_homeAuto['Device_Type'] = 'Home Automation'
merged_homeAuto['Device_number'] = 3
merged_homeAuto.to_csv('merged_homeAuto.csv', index=False)

merged_homeAuto
```

Figure 07: **Producing 'merged_homeAuto' dataset**

	Unnamed: 0	L4_tcp	L4_udp	L7_http	L7_https	port_class_src	port_class_dst	pck_size	ip_dst_new	ethernet_frame_size	...	kurt_e	var_e
0	0	1	0	0	0	2	2	313	3232270655	347	...	-1.500000	14501.555556
1	1	1	0	0	0	2	2	20	875292917	54	...	-1.593996	15458.687500
2	2	1	0	0	0	2	2	233	875292917	267	...	-1.609452	13833.840000
3	3	1	0	0	0	2	2	89	3232270655	123	...	-1.466325	12159.138889
4	4	1	0	0	0	2	2	20	875292917	54	...	-1.371188	12340.489796
...	...	...	...	...	...	...	...	...	...	...	...	...	...
19803	1	1	0	0	0	2	3	89	3232270704	123	...	0.341219	4794.385802
19804	2	1	0	0	0	3	2	20	1679138039	54	...	-0.139765	5490.847645
19805	3	0	1	0	0	3	2	180	4294967295	214	...	-0.005610	5257.127500
19806	4	0	1	0	0	3	2	180	4294967295	214	...	-3.000000	0.000000
19807	5	0	1	0	0	3	2	180	4294967295	214	...	-3.000000	0.000000

19808 rows × 51 columns

Figure 08: **'merged_homeAuto' dataset (I)**

q3_e	q1_e	iqr_e	epoch_timestamp	inter_arrival_time	time_since_previously_displayed_frame	Device_Type	Device_number
254.50	108.00	146.50	1.633634e+09	0.000000	0.000000	Home Automation	3
225.25	20.00	205.25	1.633634e+09	0.016975	0.016975	Home Automation	3
233.00	20.00	213.00	1.633634e+09	0.028957	0.045932	Home Automation	3
223.75	37.25	186.50	1.633634e+09	0.086827	0.132759	Home Automation	3
214.50	20.00	194.50	1.633634e+09	0.133683	0.266442	Home Automation	3
...	...	...	...	...	...	...	...
180.00	111.75	68.25	1.633374e+09	0.086991	0.086991	Home Automation	3
180.00	89.00	91.00	1.633374e+09	0.277293	0.364284	Home Automation	3
180.00	89.00	91.00	1.633374e+09	2.977725	3.342009	Home Automation	3
180.00	180.00	0.00	1.633374e+09	5.487765	8.829774	Home Automation	3
180.00	180.00	0.00	1.633374e+09	5.257642	14.087416	Home Automation	3

<u>Figure 09:</u> ‘merged_homeAuto’ dataset (II)

```
# Code Here (Add as many cells as you need)
merged_df = pd.concat([merged_audio, merged_camera, merged_homeAuto], ignore_index=True)

merged_df
```

<u>Figure 10:</u> Producing ‘merged_df’ dataset

	Unnamed: 0	L4_tcp	L4_udp	L7_http	L7_https	port_class_src	port_class_dst	pck_size	ip_dst_new	ethernet_frame_size	...	kurt_e	var_e
0	0	0	1	0	0	2	2	44	4026531835	78	...	0.074898	1594.240000
1	1	0	1	0	0	2	2	44	3758096634	78	...	0.858579	1336.555556
2	2	0	0	0	0	0	0	64	3232270593	98	...	0.968490	1168.489796
3	3	0	1	0	0	2	1	40	134744072	74	...	1.705787	1038.937500
4	4	0	1	0	0	1	2	56	3232270754	90	...	2.127407	926.222222
...	...	...	...	...	...	...	...	...	...	...	...	...	...
229752	1	1	0	0	0	2	3	89	3232270704	123	...	0.341219	4794.385802
229753	2	1	0	0	0	3	2	20	1679138039	54	...	-0.139765	5490.847645
229754	3	0	1	0	0	3	2	180	4294967295	214	...	-0.005610	5257.127500
229755	4	0	1	0	0	3	2	180	4294967295	214	...	-3.000000	0.000000
229756	5	0	1	0	0	3	2	180	4294967295	214	...	-3.000000	0.000000

229757 rows × 51 columns

<u>Figure 11:</u> ‘merged_df’ dataset (I)

q3_e	q1_e	iqr_e	epoch_timestamp	inter_arrival_time	time_since_previously_displayed_frame	Device_Type	Device_number
44.0	32.00	12.00	1.634054e+09	0.000000	0.000000	Audio	1
44.0	32.00	12.00	1.634054e+09	0.000000	0.000000	Audio	1
54.0	32.00	22.00	1.634054e+09	0.025100	0.025100	Audio	1
49.0	32.00	17.00	1.634054e+09	0.740150	0.765250	Audio	1
56.0	32.00	24.00	1.634054e+09	0.025906	0.791156	Audio	1
...	...	...	...	...	...	...	...
180.0	111.75	68.25	1.633374e+09	0.086991	0.086991	Home Automation	3
180.0	89.00	91.00	1.633374e+09	0.277293	0.364284	Home Automation	3
180.0	89.00	91.00	1.633374e+09	2.977725	3.342009	Home Automation	3
180.0	180.00	0.00	1.633374e+09	5.487765	8.829774	Home Automation	3
180.0	180.00	0.00	1.633374e+09	5.257642	14.087416	Home Automation	3

<u>Figure 12:</u> ‘merged_df’ dataset (II)

Next, We performed Exploratory Data Analysis (EDA) on our final dataset 'merged_df' and continued further analysis.

Using 'merged_df.info()' we got an overview of the Features, Range, Non-Null value counts, Memory Usage, and Data Types.

```
# Overview about Columns, Range, Non-Null Value Counts, Memory Usage and Data Types (#df.dtpyes)
# Code Here
merged_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 229757 entries, 0 to 229756
Data columns (total 51 columns):
 #   Column                                Non-Null Count   Dtype
---  ------                                --------------   -----
 0   Unnamed: 0                            229757 non-null  int64
 1   L4_tcp                                229757 non-null  int64
 2   L4_udp                                229757 non-null  int64
 3   L7_http                               229757 non-null  int64
 4   L7_https                              229757 non-null  int64
 5   port_class_src                        229757 non-null  int64
 6   port_class_dst                        229757 non-null  int64
 7   pck_size                              229757 non-null  int64
 8   ip_dst_new                            229757 non-null  int64
 9   ethernet_frame_size                   229757 non-null  int64
 10  ttl                                   229757 non-null  int64
 11  total_length                          229757 non-null  int64
 12  protocol                              229757 non-null  int64
 13  source_port                           229757 non-null  int64
 14  dest_port                             229757 non-null  int64
 15  DNS_count                             229757 non-null  int64
 16  NTP_count                             229757 non-null  int64
 17  ARP_count                             229757 non-null  int64
```

<u>Figure 13:</u> 'merged_df.info()' (I)

```
18  cnt                                     229757 non-null  int64
19  L3_ip_dst_count                         229757 non-null  int64
20  most_freq_d_ip                          229757 non-null  int64
21  most_freq_prot                          229757 non-null  int64
22  most_freq_sport                         229757 non-null  int64
23  most_freq_dport                         229757 non-null  int64
24  sum_et                                  229757 non-null  int64
25  min_et                                  229757 non-null  int64
26  max_et                                  229757 non-null  int64
27  med_et                                  229757 non-null  float64
28  average_et                              229757 non-null  float64
29  skew_et                                 229757 non-null  float64
30  kurt_et                                 229757 non-null  float64
31  var                                     229757 non-null  float64
32  q3                                      229757 non-null  float64
33  q1                                      229757 non-null  float64
34  iqr                                     229757 non-null  float64
35  sum_e                                   229757 non-null  int64
36  min_e                                   229757 non-null  int64
37  max_e                                   229757 non-null  int64
38  med                                     229757 non-null  float64
39  average                                 229757 non-null  float64
40  skew_e                                  229757 non-null  float64
41  kurt_e                                  229757 non-null  float64

 42  var_e                                   229757 non-null  float64
 43  q3_e                                    229757 non-null  float64
 44  q1_e                                    229757 non-null  float64
 45  iqr_e                                   229757 non-null  float64
 46  epoch_timestamp                         229757 non-null  float64
 47  inter_arrival_time                      229757 non-null  float64
 48  time_since_previously_displayed_frame   229757 non-null  float64
 49  Device_Type                             229757 non-null  object
 50  Device_number                           229757 non-null  int64
dtypes: float64(19), int64(31), object(1)
memory usage: 89.4+ MB
```

<u>Figure 14:</u> ‘merged_df.info()’ (II)

Using ‘merged_df.columns’ we got name of all the features in our dataset.

```
#Show All Column Names
print(merged_df.columns)
```

```
Index(['Unnamed: 0', 'L4_tcp', 'L4_udp', 'L7_http', 'L7_https',
       'port_class_src', 'port_class_dst', 'pck_size', 'ip_dst_new',
       'ethernet_frame_size ', 'ttl', 'total_length', 'protocol ',
       'source_port', 'dest_port', 'DNS_count', 'NTP_count', 'ARP_count',
       'cnt', 'L3_ip_dst_count', 'most_freq_d_ip', 'most_freq_prot',
       'most_freq_sport', 'most_freq_dport', 'sum_et', 'min_et', 'max_et',
       'med_et', 'average_et', 'skew_et', 'kurt_et', 'var', 'q3', 'q1', 'iqr',
       'sum_e', 'min_e', 'max_e', 'med', 'average', 'skew_e', 'kurt_e',
       'var_e', 'q3_e', 'q1_e', 'iqr_e', 'epoch_timestamp',
       'inter_arrival_time', 'time_since_previously_displayed_frame',
       'Device_Type', 'Device_number'],
      dtype='object')
```

<u>Figure 15:</u> ‘merged_columns’

```
# Count rows and colums
# Code Here
merged_rows, merged_cols = merged_df.shape
print(f"The dataset has {merged_rows} rows and {merged_cols} columns.")
```

```
The dataset has 229757 rows and 51 columns.
```

```
# Total cells in the dataset
# Code Here
merged_cells = merged_rows * merged_cols
print(f"The dataset has Total {merged_cells} num of cells")
```

```
The dataset has Total 11717607 num of cells
```

<u>Figure 16:</u> Total row, columns, and total cells in ‘merged_df’

After that, We begin with the Data Cleaning process.

Using ‘merged_df.duplicated()’ we check for any duplicated values. In our dataset, there were not any duplicated values.

```
# Code Here (Add as many cells as you need)
merged_df.duplicated()
```

```
0         False
1         False
2         False
3         False
4         False
          ...
229752    False
229753    False
229754    False
229755    False
229756    False
Length: 229757, dtype: bool
```

<u>Figure 17:</u> Identifying duplicate values

Using ‘merged_df.isnull().sum()’ we check for any missing values. In our dataset, there were not any missing values.

```
# Code Here (Add as many cells as you need)
print(merged_df.isnull().sum())
```

```
Unnamed: 0                                0
L4_tcp                                    0
L4_udp                                    0
L7_http                                   0
L7_https                                  0
port_class_src                            0
port_class_dst                            0
pck_size                                  0
ip_dst_new                                0
ethernet_frame_size                       0
ttl                                       0
total_length                              0
protocol                                  0
source_port                               0
dest_port                                 0
DNS_count                                 0
NTP_count                                 0
ARP_count                                 0
cnt                                       0
L3_ip_dst_count                           0
most_freq_d_ip                            0
most_freq_prot                            0
most_freq_sport                           0
most_freq_dport                           0
sum_et                                    0
```

<u>Figure 18:</u> Identifying missing values (I)

```
min_et                                    0
max_et                                    0
med_et                                    0
average_et                                0
skew_et                                   0
kurt_et                                   0
var                                       0
q3                                        0
q1                                        0
iqr                                       0
sum_e                                     0
min_e                                     0
max_e                                     0
med                                       0
average                                   0
skew_e                                    0
kurt_e                                    0
var_e                                     0
q3_e                                      0
q1_e                                      0
iqr_e                                     0
epoch_timestamp                           0
inter_arrival_time                        0
time_since_previously_displayed_frame     0
Device_Type                               0
Device_number                             0
dtype: int64
```

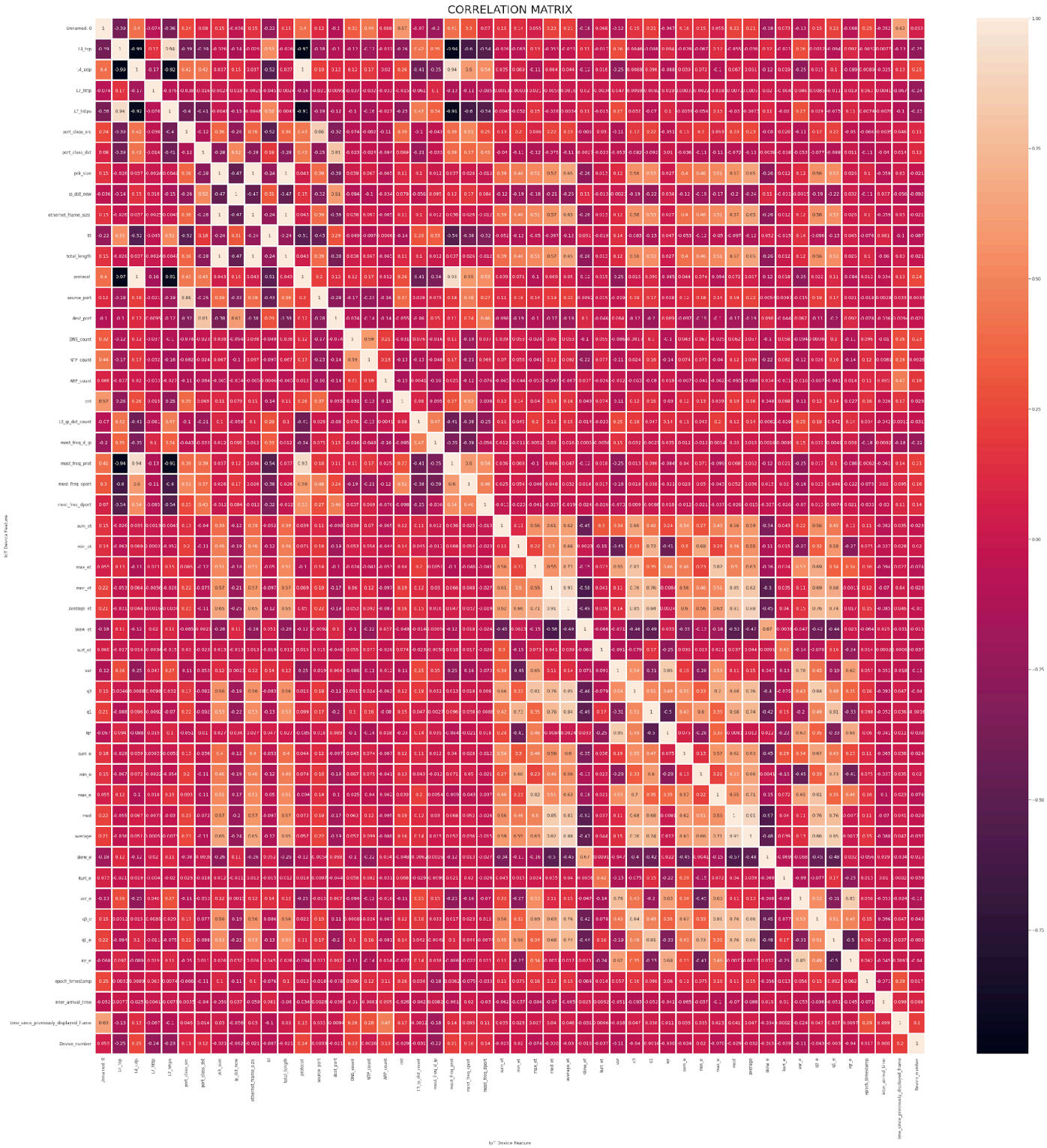
<u>Figure 19:</u> Identifying missing values (II)

Using 'merged_df.corr()', we visualized the column correlation of the dataset and then plot Heat Map.

```
# Code Here (Add as many cells as you need)

correlation_B = merged_df.corr()
correlation_B
```

<u>Figure 20:</u> Column correlation


CORRELATION MATRIX
IoT Device Feature